

Acoustic radiation by structures

MG3416–Advanced Structural Acoustics - Lecture #7

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Navier-Stokes equations

- The mass, momentum, and energy conservation equations for a fluid flow (ignoring input mass, momentum and heat) read:

$$\left\{ \begin{array}{l} \frac{d\rho}{dt} = -\rho \nabla \cdot \mathbf{v} , \\ \rho \frac{d\mathbf{v}}{dt} = \nabla \cdot \boldsymbol{\sigma} , \\ \rho T \frac{ds}{dt} = \boldsymbol{\tau} : \nabla \mathbf{v} - \nabla \cdot \mathbf{q} , \end{array} \right.$$

where $\frac{d}{dt} \equiv \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla$ and:

- $\mathbf{v}$ is the fluid velocity, ρ the density, T the temperature, s the (specific) entropy;
- $\boldsymbol{\sigma} = -p\mathbf{I} + \boldsymbol{\tau}$ is the stress tensor, p is the (static) fluid pressure, $\boldsymbol{\tau}$ is the viscous stress tensor;
- $\mathbf{q}$ is the heat flux vector.

Euler equations

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- **Ideal fluid:** $\boldsymbol{\tau} = \mathbf{0}$, $\mathbf{q} = \mathbf{0}$, and the conservation equations then read:

$$\begin{cases} \frac{d\rho}{dt} = -\rho \nabla \cdot \mathbf{v}, \\ \frac{d\mathbf{v}}{dt} = -\frac{1}{\rho} \nabla p, \\ \frac{ds}{dt} = 0. \end{cases}$$

- The flow is **isentropic** (each fluid particle has constant entropy), and by the equation of state $p = p(\rho, s)$:

$$\frac{dp}{dt} = c^2 \frac{d\rho}{dt}, \quad c^2(\rho, s) = \left. \frac{\partial p}{\partial \rho} \right|_s.$$

Linearized Euler equations

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- Linearization about a stationary fluid flow $(\varrho_0, \mathbf{v}_0, p_0)$:

$$\begin{cases} (\mathbf{v}_0 \cdot \nabla) \varrho_0 = -\varrho_0 \nabla \cdot \mathbf{v}_0, \\ (\mathbf{v}_0 \cdot \nabla) \mathbf{v}_0 = -\frac{1}{\varrho_0} \nabla p_0, \\ (\mathbf{v}_0 \cdot \nabla) p_0 = c_0^2 (\mathbf{v}_0 \cdot \nabla) \varrho_0. \end{cases}$$

- The actual flow is a perturbation $(\varrho', \mathbf{v}', p')$ of the stationary flow:

$$\begin{cases} \varrho(\mathbf{x}, t) = \varrho_0(\mathbf{x}) + \varrho'(\mathbf{x}, t), \\ \mathbf{v}(\mathbf{x}, t) = \mathbf{v}_0(\mathbf{x}) + \mathbf{v}'(\mathbf{x}, t), \\ p(\mathbf{x}, t) = p_0(\mathbf{x}) + p'(\mathbf{x}, t). \end{cases}$$

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- Let the stationary flow be at rest $\mathbf{v}_0 = \mathbf{0}$ in the absence of perturbations (waves), the **linearized Euler equations** finally yields:

$$\begin{cases} \frac{\partial \varrho'}{\partial t} = -\varrho_0 \nabla \cdot \mathbf{v}' \\ \frac{\partial \mathbf{v}'}{\partial t} = -\frac{1}{\varrho_0} \nabla p' \\ \frac{\partial p'}{\partial t} = c_0^2 \left(\frac{\partial \varrho'}{\partial t} + \mathbf{v}' \cdot \nabla \varrho_0 \right) . \end{cases}$$

- Hence the **acoustic wave equation** reads:

$$\boxed{\frac{1}{c_0^2} \frac{\partial^2 p'}{\partial t^2} - \varrho_0 \nabla \cdot \left(\frac{1}{\varrho_0} \nabla p' \right) = 0} .$$

Green's function

- The **Green's function** $\mathcal{G}(\mathbf{x}, t; \mathbf{x}_0, t_0)$ is the solution of:

$$\left(\frac{1}{c_0^2} \frac{\partial^2}{\partial t^2} - \varrho_0 \nabla_{\mathbf{x}} \cdot \frac{1}{\varrho_0} \nabla_{\mathbf{x}} \right) \mathcal{G}(\mathbf{x}, t; \mathbf{x}_0, t_0) = \delta(t - t_0) \delta(\mathbf{x} - \mathbf{x}_0),$$
$$\mathcal{G}(\mathbf{x}, t; \mathbf{x}_0, t_0) = 0, \quad t < t_0,$$

for all $(\mathbf{x}, t) \in \mathbb{R}^3 \times \mathbb{R}$ and some $(\mathbf{x}_0, t_0) \in \mathbb{R}^3 \times \mathbb{R}$ fixed.

- The solution in homogeneous space is:

$$\begin{aligned} \mathcal{G}_0(\mathbf{x}, t; \mathbf{x}_0, t_0) &= G_0(\mathbf{x} - \mathbf{x}_0, t - t_0) \\ &= \frac{1}{4\pi \|\mathbf{x} - \mathbf{x}_0\|} \delta \left(t - t_0 - \frac{\|\mathbf{x} - \mathbf{x}_0\|}{c_0} \right). \end{aligned}$$

- **Remarks:** $\mathcal{G}_0$ is singular at $\mathbf{x} = \mathbf{x}_0$, and $\mathcal{G}_0 \equiv 0$ if $t_0 > t$ by **causality** (one cannot observe a signal before it is emitted).

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- The acoustic wave equation in an open domain Ω with boundary $\partial\Omega = \Gamma$, sources f , and initial conditions $p(\mathbf{x}, t_0) = p_0(\mathbf{x})$, $\partial_t p(\mathbf{x}, t_0) = q_0(\mathbf{x})$:

$$\left(\frac{1}{c_0^2} \frac{\partial^2}{\partial t^2} - \varrho_0 \nabla \cdot \frac{1}{\varrho_0} \nabla \right) p(\mathbf{x}, t) = f(\mathbf{x}, t), \quad (\mathbf{x}, t) \in \Omega \times \mathbb{R}.$$

- Then its solution reads for all $(\mathbf{x}, t) \in \Omega \times \mathbb{R}$ —the **Kirchhoff's theorem** (assume $\varrho_0 \simeq C^{\text{st}}$):

$$\begin{aligned} p(\mathbf{x}, t) = & \int_{t_0}^t \int_{\Omega} \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) f(\mathbf{y}, \tau) d\mathbf{y} d\tau \\ & + \int_{t_0}^t \int_{\Gamma} \left(\mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) \frac{\partial p(\mathbf{y}, \tau)}{\partial \mathbf{n}_{\mathbf{y}}} - \frac{\partial \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau)}{\partial \mathbf{n}_{\mathbf{y}}} p(\mathbf{y}, \tau) \right) dS_{\mathbf{y}} d\tau \\ & + \int_{\Omega} \frac{1}{c_0^2} \left(\mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, t_0) q_0(\mathbf{y}) - \frac{\partial \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau)}{\partial \tau} \Big|_{\tau=t_0} p_0(\mathbf{y}) \right) d\mathbf{y}. \end{aligned}$$

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- **Proof:** multiplying the wave equation for $p(\mathbf{y}, \tau)$ by $\mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau)$ and integrating on $\Omega \times [t_0, t_1]$ yields

$$\begin{aligned} 0 &= \int_{t_0}^{t_1} \int_{\Omega} \left(\frac{1}{c_0^2} \frac{\partial^2 p(\mathbf{y}, \tau)}{\partial \tau^2} - \Delta_{\mathbf{y}} p(\mathbf{y}, \tau) - f(\mathbf{y}, \tau) \right) \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) \, d\mathbf{y} d\tau \\ &= \int_{t_0}^{t_1} \int_{\Omega} \frac{1}{c_0^2} \left[\frac{\partial}{\partial \tau} \left(\mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) \frac{\partial p(\mathbf{y}, \tau)}{\partial \tau} \right) - \frac{\partial \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau)}{\partial \tau} \frac{\partial p(\mathbf{y}, \tau)}{\partial \tau} \right] \, d\mathbf{y} d\tau \\ &\quad - \int_{t_0}^{t_1} \int_{\Omega} [\nabla_{\mathbf{y}} \cdot (\mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) \nabla_{\mathbf{y}} p(\mathbf{y}, \tau)) \\ &\quad \quad \quad - \nabla_{\mathbf{y}} \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) \cdot \nabla_{\mathbf{y}} p(\mathbf{y}, \tau)] \, d\mathbf{y} d\tau \\ &\quad - \int_{t_0}^{t_1} \int_{\Omega} \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) f(\mathbf{y}, \tau) \, d\mathbf{y} d\tau \end{aligned}$$

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- **Proof:** multiplying the wave equation for $p(\mathbf{y}, \tau)$ by $\mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau)$ and integrating on $\Omega \times [t_0, t_1]$ yields

$$\begin{aligned} 0 &= \int_{t_0}^{t_1} \int_{\Omega} \frac{1}{c_0^2} \frac{\partial}{\partial \tau} \left(\mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) \frac{\partial p(\mathbf{y}, \tau)}{\partial \tau} - \frac{\partial \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau)}{\partial \tau} p(\mathbf{y}, \tau) \right) d\mathbf{y} d\tau \\ &\quad - \int_{t_0}^{t_1} \int_{\Omega} \nabla_{\mathbf{y}} \cdot (\mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) \nabla_{\mathbf{y}} p(\mathbf{y}, \tau) - p(\mathbf{y}, \tau) \nabla_{\mathbf{y}} \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau)) d\mathbf{y} d\tau \\ &\quad + \int_{t_0}^{t_1} \int_{\Omega} \left(\frac{1}{c_0^2} \frac{\partial^2 \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau)}{\partial \tau^2} - \Delta_{\mathbf{y}} \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) \right) p(\mathbf{y}, \tau) d\mathbf{y} d\tau \\ &\quad - \int_{t_0}^{t_1} \int_{\Omega} \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) f(\mathbf{y}, \tau) d\mathbf{y} d\tau. \end{aligned}$$

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- **Proof** (cont'd): but $\mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) = \mathcal{G}_0(\mathbf{y}, t; \mathbf{x}, \tau)$ (**reciprocity**) and applying the divergence theorem yields

$$\begin{aligned} p(\mathbf{x}, t) = & \int_{\Omega} \frac{1}{c_0^2} \left[\frac{\partial \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau)}{\partial \tau} p(\mathbf{y}, \tau) - \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) \frac{\partial p(\mathbf{y}, \tau)}{\partial \tau} \right]_{\tau=t_0}^{\tau=t_1} d\mathbf{y} \\ & + \int_{t_0}^{t_1} \int_{\partial\Omega} \left(\mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) \frac{p(\mathbf{y}, \tau)}{\partial \mathbf{n}_{\mathbf{y}}} - \frac{\partial \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau)}{\partial \mathbf{n}_{\mathbf{y}}} p(\mathbf{y}, \tau) \right) d\mathbf{y} d\tau \\ & + \int_{t_0}^{t_1} \int_{\Omega} \mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, \tau) f(\mathbf{y}, \tau) d\mathbf{y} d\tau. \end{aligned}$$

- Choosing $t_1 = t^+$ one has $\mathcal{G}_0(\mathbf{x}, t; \mathbf{y}, t^+) = 0$ by causality, and obtains therefore the claimed result $\square$

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- The **Helmholtz equation** in an open domain Ω with boundary $\partial\Omega = \Gamma$ and sources $\hat{f}$:

$$-\left(\frac{\omega^2}{c^2(\mathbf{x})} + \Delta\right) \hat{p}(\mathbf{x}, \omega) = \hat{f}(\mathbf{x}, \omega), \quad (\mathbf{x}, \omega) \in \Omega \times \mathbb{R}.$$

- The **Green's function** $\hat{\mathcal{G}}(\mathbf{x}, \mathbf{x}_0, \omega)$ is the solution of:

$$-\left(\frac{\omega^2}{c^2(\mathbf{x})} + \Delta\right) \hat{\mathcal{G}}(\mathbf{x}, \mathbf{x}_0, \omega) = \delta(\mathbf{x} - \mathbf{x}_0), \quad (\mathbf{x}, \omega) \in \mathbb{R}^3 \times \mathbb{R},$$

for some $\mathbf{x}_0 \in \mathbb{R}^3$ fixed, satisfying the **Sommerfeld radiation condition** (with $c(\mathbf{x}) = c_0$ at infinity):

$$\lim_{\|\mathbf{x}\| \rightarrow \infty} \|\mathbf{x}\| \left(\frac{\mathbf{x}}{\|\mathbf{x}\|} \cdot \nabla_{\mathbf{x}} - i \frac{\omega}{c_0} \right) \hat{\mathcal{G}}(\mathbf{x}, \mathbf{x}_0, \omega) = 0.$$

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- The Green's function $\hat{\mathcal{G}}_0(\mathbf{x}, \mathbf{x}_0, \omega)$ in homogeneous space $c(\mathbf{x}) = c_0$ reads:

$$\hat{\mathcal{G}}_0(\mathbf{x}, \mathbf{x}_0, \omega) = \hat{G}_0(\mathbf{x} - \mathbf{x}_0, \omega) = \frac{e^{i \frac{\omega}{c_0} \|\mathbf{x} - \mathbf{x}_0\|}}{4\pi \|\mathbf{x} - \mathbf{x}_0\|}.$$

- Then the solution of the Helmholtz equation reads for all $(\mathbf{x}, \omega) \in \Omega \times \mathbb{R}$:

$$\begin{aligned} \hat{p}(\mathbf{x}, \omega) = & \int_{\Omega} \hat{\mathcal{G}}_0(\mathbf{x}, \mathbf{y}, \omega) \hat{f}(\mathbf{y}, \omega) d\mathbf{y} \\ & + \int_{\Gamma} \left(\hat{\mathcal{G}}_0(\mathbf{x}, \mathbf{y}, \omega) \frac{\partial \hat{p}(\mathbf{y}, \omega)}{\partial \mathbf{n}_{\mathbf{y}}} - \frac{\partial \hat{\mathcal{G}}_0(\mathbf{x}, \mathbf{y}, \omega)}{\partial \mathbf{n}_{\mathbf{y}}} \hat{p}(\mathbf{y}, \omega) \right) dS_{\mathbf{y}}. \end{aligned}$$

- **Proof:** multiply the Helmholtz equation for $\hat{p}(\mathbf{y}, \omega)$ by $\hat{\mathcal{G}}_0(\mathbf{x}, \mathbf{y}, \omega)$ and integrate on Ω .

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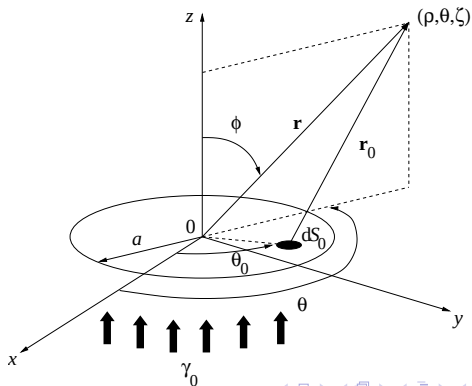
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- Computation of the acoustic field radiated by a planar surface S_0 at $z = 0$ of which normal speed is $i\omega v_0(\mathbf{x}_0, t) = \gamma_0(\mathbf{x}_0)e^{-i\omega t}$, $\mathbf{x}_0 \in S_0$.
- No source in the half-space $\{z > 0\}$.



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- Taking the normal trace of the Euler equation yields $\varrho_0 \gamma_0 = \frac{\partial \hat{p}}{\partial \mathbf{n}}$, and by Kirchhoff theorem:

$$\hat{p}(\mathbf{r}, \omega) = \int_{S_0} \left(\varrho_0 \gamma_0 \hat{\mathcal{G}}_0(\mathbf{r}, \mathbf{x}_0, \omega) - \frac{\partial \hat{\mathcal{G}}_0(\mathbf{r}, \mathbf{x}_0, \omega)}{\partial \mathbf{n}_0} \hat{p}(\mathbf{x}_0, \omega) \right) dS(\mathbf{x}_0),$$

but $\hat{p}(\mathbf{x}_0, \omega)$ remains unknown on S_0 .

- We rather consider the Green's function of the half-space $\{z > 0\}$ with rigid wall at $z = 0$, which is given by the method of images for a source $\mathbf{r}_0 = (\mathbf{x}_0, z_0 > 0)$ and its image $\mathbf{r}'_0 = (\mathbf{x}_0, -z_0)$:

$$\hat{\mathcal{G}}_+(\mathbf{r}, \mathbf{r}_0, \omega) = \frac{e^{i \frac{\omega}{c_0} \|\mathbf{r} - \mathbf{r}_0\|}}{4\pi \|\mathbf{r} - \mathbf{r}_0\|} + \frac{e^{i \frac{\omega}{c_0} \|\mathbf{r} - \mathbf{r}'_0\|}}{4\pi \|\mathbf{r} - \mathbf{r}'_0\|}.$$

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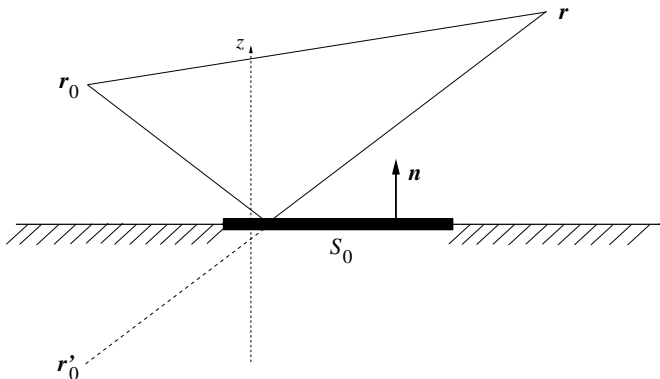
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$$\hat{\mathcal{G}}_+(\mathbf{r}, \mathbf{r}_0, \omega) = \frac{e^{i \frac{\omega}{c_0} \|\mathbf{r} - \mathbf{r}_0\|}}{4\pi \|\mathbf{r} - \mathbf{r}_0\|} + \frac{e^{i \frac{\omega}{c_0} \|\mathbf{r} - \mathbf{r}'_0\|}}{4\pi \|\mathbf{r} - \mathbf{r}'_0\|}$$

Rayleigh's integral

- Letting $z_0 \rightarrow 0$ such that $\mathbf{r}_0 \rightarrow \mathbf{x}_0$ one ends up with:

$$\hat{\mathcal{G}}_+(\mathbf{r}, \mathbf{x}_0, \omega) = \frac{e^{i\frac{\omega}{c_0}\|\mathbf{r}-\mathbf{x}_0\|}}{2\pi\|\mathbf{r}-\mathbf{x}_0\|},$$

$$\frac{\partial \hat{\mathcal{G}}_+(\mathbf{r}, \mathbf{x}_0, \omega)}{\partial n_0} = 0.$$

- Using the foregoing half-space Green's function, the pressure field radiated by the rigid surface is finally given by [Rayleigh's integral](#):

$$\hat{p}(\mathbf{r}, \omega) = \frac{\varrho_0}{2\pi} \int_{S_0} \frac{e^{i\frac{\omega}{c_0}\|\mathbf{r}-\mathbf{x}_0\|}}{\|\mathbf{r}-\mathbf{x}_0\|} \gamma_0(\mathbf{x}_0) dS(\mathbf{x}_0).$$

- **Example:** baffled piston (see tutorial class #3).

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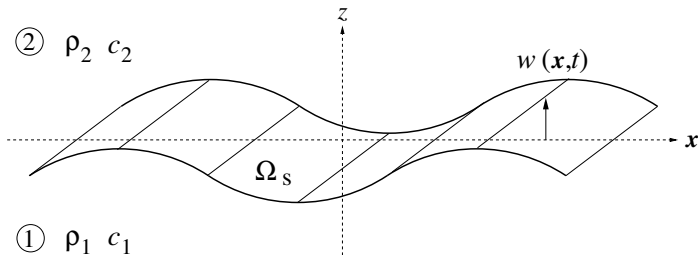
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- Coupling of a thin plate with the surrounding air or water.
- **Examples:** musical instruments, underwater acoustics of submarines, sound insulation of buildings, architectural acoustics...
- For hints see for instance

http://vibroacoustique.fr/cours/M01_C01/co/VAC_M01_C01_web.html.



Infinite plate *in vacuo*

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- The equation of motion for the normal displacement $w(\mathbf{x}, t)$ of an infinite thin plate $\Omega_s = \{\mathbf{x} \in \mathbb{R}^2\}$ using [Kirchhoff-Love kinematics](#):

$$\varrho_s \partial_t^2 w + D \Delta_{\mathbf{x}}^2 w = \mathbf{f}_{\text{ext}} \cdot \mathbf{e}_z ,$$

where $D = \frac{Eh^3}{12(1-\nu^2)}$, h is the plate thickness, E is Young's modulus, ν is Poisson's ratio, ϱ is the density, $\varrho_s = \varrho h$, and $\Delta_{\mathbf{x}}^2 = (\nabla_{\mathbf{x}} \cdot \nabla_{\mathbf{x}})^2$.

- Thin plate assumption $h \ll \lambda$ (see [Mindlin kinematics](#) for thick plates—the like of Timoshenko theory for thick beams).

Plane wave solution and dispersion equation

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- Plane wave solution:

$$w(\mathbf{x}, t) = W e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \quad \mathbf{k} \in \mathbb{R}^2, \omega \in \mathbb{R}.$$

- Plugging-in this *ansatz* in the homogeneous ($\mathbf{f}_{\text{ext}} \equiv \mathbf{0}$) equation of motion yields the dispersion equation and the flexural wave number $\omega \mapsto k_b(\omega)$:

$$-\varrho_s \omega^2 + D \|\mathbf{k}\|^4 = 0, \quad k_b(\omega) = \sqrt[4]{\frac{\varrho_s \omega^2}{D}}.$$

- The phase velocity $c_b := \frac{\omega}{k_b}$ and group velocity and $c_g := \frac{d\omega}{dk}$ are:

$$c_b(\omega) = \sqrt[4]{\frac{D \omega^2}{\varrho_s}}, \quad c_g(\omega) = 2 c_b(\omega).$$

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- **Remark** (group velocity): Consider the wave trains $\cos(\omega t - kx)$ and $\cos[(\omega + d\omega)t - (k + dk)x]$, their sum is:

$$2 \cos\left(\frac{d\omega}{2}t - \frac{dk}{2}x\right) \cos\left[\left(\omega + \frac{d\omega}{2}\right)t - \left(k + \frac{dk}{2}\right)x\right].$$

- The average wave front is located at the position x and time t s.t. $td\omega - xdk = 0$, thus it travels at the group velocity:

$$c_g = \frac{d\omega}{dk}.$$

- Group velocity = speed of transport of the energy.
- Euler-Bernoulli beam: $c_b(\omega) = \sqrt[4]{\frac{EI\omega^2}{\rho A}}$, $c_g(\omega) = 2c_b(\omega)$.

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- For fluid media $\Omega_j = \{\mathbf{r} = (\mathbf{x}, z \lessgtr 0)\}$ above and below the plate with density and speed of sound ϱ_2, c_2 ($z > 0$) and ϱ_1, c_1 ($z < 0$), respectively, the **linearized Euler equation** (balance of momentum) in frequency domain reads:

$$i\omega\varrho_j\hat{\mathbf{v}}_j + \nabla_{\mathbf{r}}\hat{p}_j = 0, \quad j = 1, 2,$$

with the Sommerfeld radiation condition for $\|\mathbf{r}\| \rightarrow \infty$.

- The bending equation for the plate in frequency domain reads:

$$(-\varrho_s\omega^2 + D\Delta_{\mathbf{x}}^2)\hat{w}(\mathbf{x}, \omega) = \hat{p}_1(\mathbf{x}, 0^-, \omega) - \hat{p}_2(\mathbf{x}, 0^+, \omega),$$

with continuity of the normal speed:

$$i\omega\hat{w}(\mathbf{x}, \omega)|_{z=0^-} = \hat{\mathbf{v}}_1(\mathbf{x}, 0^-, \omega) \cdot \mathbf{e}_z,$$

$$i\omega\hat{w}(\mathbf{x}, \omega)|_{z=0^+} = \hat{\mathbf{v}}_2(\mathbf{x}, 0^+, \omega) \cdot \mathbf{e}_z.$$

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- The pressure fields may be sought for as:

$$\hat{p}_1(\mathbf{x}, z < 0, \omega) = P_1 e^{i(\mathbf{k}_1 \cdot \mathbf{x} - k_{1n} z)},$$

$$\hat{p}_2(\mathbf{x}, z > 0, \omega) = P_2 e^{i(\mathbf{k}_2 \cdot \mathbf{x} + k_{2n} z)},$$

owing to the Sommerfeld radiation conditions (no incoming waves), where $k_{jn}(\omega) = (\frac{\omega^2}{c_j^2} - \|\mathbf{k}_j\|^2)^{\frac{1}{2}}$.

- The plate bending motion may be sought for as:

$$\hat{w}(\mathbf{x}, \omega) = W e^{i\mathbf{k} \cdot \mathbf{x}}.$$

- Taking the normal trace of the Euler equation on either side of the plate yields $-i\omega \varrho_j \hat{\mathbf{v}}_j \cdot \mathbf{e}_z = \partial_z \hat{p}_j$, therefore:

$$\hat{p}_1(\mathbf{x}, 0^-, \omega) = \frac{i\varrho_1 \omega^2}{k_{1n}(\omega)} \hat{w}(\mathbf{x}, \omega),$$

$$\hat{p}_2(\mathbf{x}, 0^+, \omega) = -\frac{i\varrho_2 \omega^2}{k_{2n}(\omega)} \hat{w}(\mathbf{x}, \omega).$$

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- Because these relationships are valid for all $\mathbf{x} \in \mathbb{R}^2$, the **Snell-Descartes law**—conservation of the tangent wave vector $\mathbf{k} \in \mathbb{R}^2$ —is obtained:

$$\boxed{\mathbf{k}_1 = \mathbf{k} = \mathbf{k}_2},$$

and the following **dispersion equation for the plate surrounded by the fluids** holds:

$$-i\omega^2 \left(\frac{\varrho_1}{\sqrt{\frac{\omega^2}{c_1^2} - \|\mathbf{k}\|^2}} + \frac{\varrho_2}{\sqrt{\frac{\omega^2}{c_2^2} - \|\mathbf{k}\|^2}} \right) - \omega^2 \varrho_s + D \|\mathbf{k}\|^4 = 0.$$

- Consequently the pressure fields in the fluid media are:

$$\hat{p}_1(\mathbf{x}, z < 0, \omega) = \frac{i\varrho_1\omega^2}{k_{1n}(\omega)} \hat{w}(\mathbf{x}, \omega) e^{-ik_{1n}(\omega)z},$$

$$\hat{p}_2(\mathbf{x}, z > 0, \omega) = -\frac{i\varrho_2\omega^2}{k_{2n}(\omega)} \hat{w}(\mathbf{x}, \omega) e^{+ik_{2n}(\omega)z}.$$

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- Assume now $c_1 = c_2 = c_0$ and $\varrho_1 = \varrho_2 = \varrho_0$ for a light fluid (air) such that $\varepsilon = \frac{2\varrho_0}{\varrho_s k_b} \ll 1$, where $k_b^4 = \frac{\varrho_s \omega^2}{D}$.
- Let $\alpha(\omega)$ be defined by $i\alpha = k_n = (k_0^2 - \|\mathbf{k}\|^2)^{\frac{1}{2}}$ where $\mathbf{k} \in \mathbb{R}^2$ is the plate (tangent) wave vector and $k_0 = \frac{\omega}{c_0}$; then:

$$-\varepsilon \frac{k_b^5}{\alpha} - k_b^4 + (k_0^2 + \alpha^2)^2 = 0.$$

- The four roots for $\varepsilon \ll 1$ are:

$$\begin{aligned}\alpha_{1,2} &= \pm i \sqrt{k_0^2 + k_b^2}, \\ \alpha_{3,4} &= \pm \sqrt{k_b^2 - k_0^2} && \text{if } k_b^2 > k_0^2, \\ \alpha_{3,4} &= \pm i \sqrt{k_0^2 - k_b^2} && \text{if } k_b^2 < k_0^2.\end{aligned}$$

First case: $\alpha_{1,2} = \pm i\sqrt{k_0^2 + k_b^2}$

- $i\alpha_{1,2} = \mp(k_0^2 + k_b^2)^{\frac{1}{2}}$ but only the first instance is acceptable physically in order to fulfill the Sommerfeld radiation condition.
- The pressure waves within the air are **propagative**:

$$\hat{p}(\mathbf{x}, z < 0, \omega) = \frac{i\rho_0\omega^2\hat{w}(\mathbf{x}, \omega)}{\sqrt{k_0^2 + k_b^2}} e^{-iz\sqrt{k_0^2 + k_b^2}},$$

$$\hat{p}(\mathbf{x}, z > 0, \omega) = -\frac{i\rho_0\omega^2\hat{w}(\mathbf{x}, \omega)}{\sqrt{k_0^2 + k_b^2}} e^{+iz\sqrt{k_0^2 + k_b^2}}.$$

- Then $\|\mathbf{k}\|^2 = -k_b^2$ or $\|\mathbf{k}\| = \pm ik_b$, and therefore the bending waves within the plate are **evanescent**:

$$\hat{w}(\mathbf{x}, \omega) = W e^{-k_b(\omega)\|\mathbf{x}\|}.$$

Second case: $\alpha_{3,4} = \pm\sqrt{k_b^2 - k_0^2}$ for $k_b^2 > k_0^2$

- $i\alpha_{3,4} = \pm i(k_b^2 - k_0^2)^{\frac{1}{2}}$ but only the first instance is acceptable physically in order to fulfill the Sommerfeld radiation condition.
- The pressure waves within the air are **evanescent**:

$$\hat{p}(\mathbf{x}, z < 0, \omega) = \frac{\varrho_0 \omega^2 \hat{w}(\mathbf{x}, \omega)}{\sqrt{k_b^2 - k_0^2}} e^{z\sqrt{k_b^2 - k_0^2}},$$

$$\hat{p}(\mathbf{x}, z > 0, \omega) = -\frac{\varrho_0 \omega^2 \hat{w}(\mathbf{x}, \omega)}{\sqrt{k_b^2 - k_0^2}} e^{-z\sqrt{k_b^2 - k_0^2}}.$$

- Then $\|\mathbf{k}\|^2 = k_b^2$ or $\|\mathbf{k}\| = \pm k_b$, and therefore the bending waves within the plate are **propagative**:

$$\hat{w}(\mathbf{x}, \omega) = W_+ e^{+ik_b(\omega)\hat{\mathbf{k}} \cdot \mathbf{x}} + W_- e^{-ik_b(\omega)\hat{\mathbf{k}} \cdot \mathbf{x}}.$$

Third case: $\alpha_{3,4} = \pm i\sqrt{k_0^2 - k_b^2}$ for $k_b^2 < k_0^2$

- $i\alpha_{3,4} = \mp(k_0^2 - k_b^2)^{\frac{1}{2}}$ but only the first instance is acceptable physically in order to fulfill the Sommerfeld radiation condition.
- The pressure waves within the air are **propagative**:

$$\hat{p}(\mathbf{x}, z < 0, \omega) = \frac{i\rho_0\omega^2\hat{w}(\mathbf{x}, \omega)}{\sqrt{k_0^2 - k_b^2}} e^{-iz\sqrt{k_0^2 - k_b^2}},$$

$$\hat{p}(\mathbf{x}, z > 0, \omega) = -\frac{i\rho_0\omega^2\hat{w}(\mathbf{x}, \omega)}{\sqrt{k_0^2 - k_b^2}} e^{+iz\sqrt{k_0^2 - k_b^2}}.$$

- Then $\|\mathbf{k}\|^2 = k_b^2$ or $\|\mathbf{k}\| = \pm k_b$, and therefore the bending waves within the plate are **propagative**:

$$\hat{w}(\mathbf{x}, \omega) = W_+ e^{+ik_b(\omega)\hat{\mathbf{k}}\cdot\mathbf{x}} + W_- e^{-ik_b(\omega)\hat{\mathbf{k}}\cdot\mathbf{x}}.$$

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- The **critical frequency** such that $k_b(\omega) = k_0 = \frac{\omega}{c_0}$ is $\omega_c = c_0^2 \sqrt{\frac{\rho_s}{D}}$.
- The **radiation efficiency** (*facteur de rayonnement*) $\sigma(\omega)$ of the plate is defined by:

$$\sigma(\omega) = \frac{1}{\rho_0 c_0} \Re \left[\frac{\hat{p}(\mathbf{x}, 0^+, \omega)}{-i\omega \hat{w}(\mathbf{x}, \omega)} \right].$$

- For the various foregoing situations it is:

$$\sigma(\omega) = \begin{cases} \frac{1}{\sqrt{1 + \frac{\omega_c}{\omega}}} & \text{for evanescent waves in the plate,} \\ 0 & \text{for propagative waves in the plate with } \omega < \omega_c, \\ \frac{1}{\sqrt{1 - \frac{\omega_c}{\omega}}} & \text{for propagative waves in the plate with } \omega > \omega_c. \end{cases}$$

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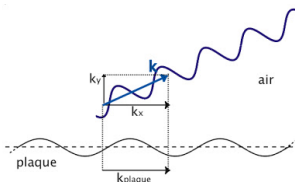
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- **Subsonic** plate waves such that $\omega < \omega_c$ or $c_b < c_0$ or $\lambda_b < \lambda_0$ do not radiate within the air: no sound transmission.
- **Supersonic** plate waves such that $\omega > \omega_c$ or $c_b > c_0$ or $\lambda_b > \lambda_0$ do radiate within the air with maximum efficiency when $\omega \simeq \omega_c$.
- This is because by **Snell-Descartes law** the tangent wave vector is constant in the transmission process, so that if the plate wavenumber k_b is larger than the fluid wavenumber k_0 there can be no transmission at all.



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- The equation of motion for the normal displacement $w(\mathbf{x}, t)$ of a finite thin plate $\Omega_s = [0, a] \times [0, b]$ using again **Kirchhoff-Love kinematics**:

$$\varrho_s \partial_t^2 w + D \Delta_{\mathbf{x}}^2 w = \mathbf{f}_{\text{ext}} \cdot \mathbf{e}_z ,$$

where $D = \frac{Eh^3}{12(1-\nu^2)}$, h is the plate thickness, E is Young's modulus, ν is Poisson's ratio, ϱ is the density, $\varrho_s = \varrho h$, and $\Delta_{\mathbf{x}}^2 = (\nabla_{\mathbf{x}} \cdot \nabla_{\mathbf{x}})^2$.

- Boundary conditions on $\partial\Omega_s = \{x = 0, a\} \cup \{y = 0, b\}$ for simply supported plate:

$$w(\mathbf{x}, t) = 0, \quad \mathbf{M}(\mathbf{x}, t)\mathbf{n} = \mathbf{0},$$

where $-\mathbf{M} = \nu D(\Delta_{\mathbf{x}} w)\mathbf{I} + (1 - \nu)D\nabla_{\mathbf{x}} \otimes \nabla_{\mathbf{x}} w$ is the bending moment tensor.

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- **Modal expansion** solution:

$$w(\mathbf{x}, t) = \sum_{\alpha} q_{\alpha}(t) \phi_{\alpha}(\mathbf{x})$$

where $\alpha = (\alpha, \beta) \in \mathbb{N}^2 \setminus \{\mathbf{0}\}$ is a 2-index and ϕ_{α} is the corresponding eigen mode shape of a simply supported thin plate (s.t. $\int_{\Omega_s} \varrho_s \phi_{\alpha}(\mathbf{x}) \phi_{\beta}(\mathbf{x}) d\mathbf{x} = \varrho_s ab \delta_{\alpha\beta}$):

$$\phi_{\alpha}(\mathbf{x}) = 2 \sin\left(\alpha\pi \frac{x}{a}\right) \sin\left(\beta\pi \frac{y}{b}\right).$$

- Plugging-in this expansion in the homogeneous ($\mathbf{f}_{\text{ext}} \equiv \mathbf{0}$) equation of motion yields the discrete **flexural wave numbers** $\mathbf{k}_{\alpha}$:

$$\|\mathbf{k}_{\alpha}\|^2 = \left(\frac{\alpha\pi}{a}\right)^2 + \left(\frac{\beta\pi}{b}\right)^2, \quad \omega_{\alpha} = \|\mathbf{k}_{\alpha}\|^2 \sqrt{\frac{D}{\varrho_s}}.$$

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- The eigen modes of the plate *in vacuo* are still a basis for the bending motion, but they get coupled when the surrounding fluids are taken into account.
- Qualitatively one has basically for air with acoustic wavelength λ_0 and bending wavelengths $\frac{\lambda_x}{2} = \frac{a}{\alpha}$, $\frac{\lambda_y}{2} = \frac{b}{\beta}$:
 - If $\omega > \omega_c$: the radiation efficiency goes to 1;
 - If $\omega < \omega_c$: the radiation efficiency is interpreted in terms of possible boundary effects.
 - If $\lambda_x < \frac{\lambda_0}{2}$ and $\lambda_y < \frac{\lambda_0}{2}$: **corner radiation** whereby modal quadrupoles and dipoles have low radiation efficiency;
 - If $\lambda_x < \frac{\lambda_0}{2}$ or $\lambda_y < \frac{\lambda_0}{2}$: **boundary radiation**;
 - If $\lambda_x > \frac{\lambda_0}{2}$ and $\lambda_y > \frac{\lambda_0}{2}$: overall plater radiation.

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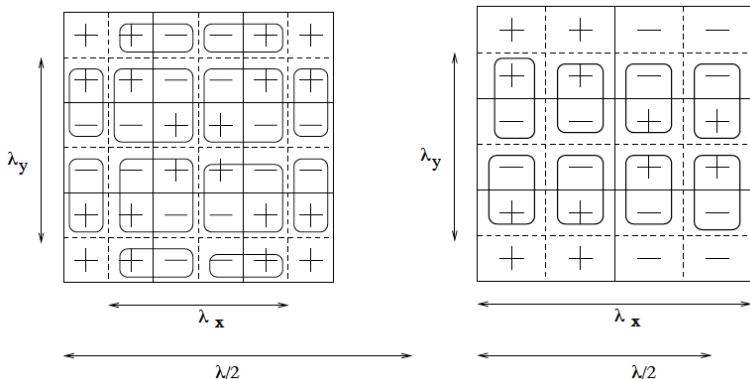
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Corner radiation (left) *vs.* boundary radiation (right).

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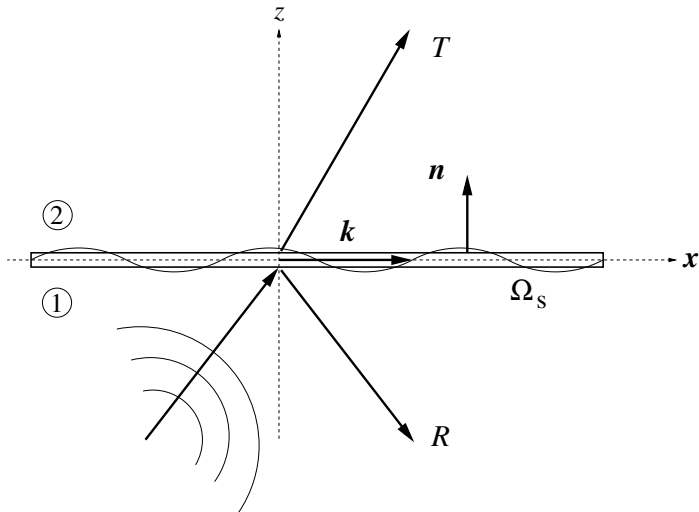
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- Fluid medium ①: incoming plane wave + reflected plane wave

$$\hat{p}_1(\mathbf{x}, z < 0, \omega) = P e^{i\mathbf{k} \cdot \mathbf{x}} \left(e^{ik_{1n}z} + R e^{-ik_{1n}z} \right).$$

- Fluid medium ②: transmitted plane wave

$$\hat{p}_2(\mathbf{x}, z > 0, \omega) = P T e^{i(\mathbf{k} \cdot \mathbf{x} + k_{2n}z)}.$$

- R and T are the amplitude **reflection coefficients**, and $\mathbf{k}$ is the tangent wave vector.
- The wave number in fluid medium ② is $k_2 = \frac{\omega}{c_2}$ and:

$$k_{2n}(\mathbf{k}) = k_2 \sqrt{1 - \left(\frac{\|\mathbf{k}\|}{k_2} \right)^2},$$

whereby transmission holds provided that $\|\mathbf{k}\| < k_2$ (**total reflection** otherwise).

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- Plate bending motion $\hat{w}(\mathbf{x}, \omega) = W e^{i\mathbf{k} \cdot \mathbf{x}}$ such that:

$$(-\varrho_s \omega^2 + D \|\mathbf{k}\|^4) \hat{w}(\mathbf{x}, \omega) = \hat{p}_1(\mathbf{x}, 0^-, \omega) - \hat{p}_2(\mathbf{x}, 0^+, \omega),$$

with boundary conditions (continuity of the normal speed $\varrho_j \omega^2 \hat{w} = \partial_z \hat{p}_j|_{z=\pm 0}$):

$$\varrho_1 \omega^2 \hat{w}(\mathbf{x}, \omega) = i k_{1n}(\mathbf{k}) P (1 - R) e^{i\mathbf{k} \cdot \mathbf{x}},$$

$$\varrho_2 \omega^2 \hat{w}(\mathbf{x}, \omega) = i k_{2n}(\mathbf{k}) P T e^{i\mathbf{k} \cdot \mathbf{x}}.$$

- One ends up with the system:

$$(-\varrho_s \omega^2 + D \|\mathbf{k}\|^4) W = P (1 + R - T),$$

$$\varrho_1 \omega^2 W = i k_{1n}(\mathbf{k}) P (1 - R),$$

$$\varrho_2 \omega^2 W = i k_{2n}(\mathbf{k}) P T.$$

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- Its solution reads:

$$\frac{W}{P} = \frac{2}{-\omega^2 \left(\varrho_s + \frac{i\varrho_1}{k_{1n}(\mathbf{k})} + \frac{i\varrho_2}{k_{2n}(\mathbf{k})} \right) + D\|\mathbf{k}\|^4},$$

$$R = 1 + \frac{i\varrho_1\omega^2}{k_{1n}(\mathbf{k})} \left(\frac{W}{P} \right),$$

$$T = -\frac{i\varrho_2\omega^2}{k_{2n}(\mathbf{k})} \left(\frac{W}{P} \right).$$

- **Remark:** this result holds for all plates for which $-\omega^2 \mathcal{Z}_s(\omega) \hat{w}(\mathbf{x}, \omega) = \hat{f}(\omega)$; consequently:

$$\frac{W}{P} = -2\omega^{-2} \left(\mathcal{Z}_s(\omega) + \frac{i\varrho_1}{k_{1n}(\mathbf{k})} + \frac{i\varrho_2}{k_{2n}(\mathbf{k})} \right)^{-1}.$$

Acoustic transparency

- The **acoustic transparency** or **transmissivity** is:

$$\tau(\omega) = \frac{\Pi_T(\omega)}{\Pi_{IN}(\omega)},$$

where Π_{IN} is the **incident power**:

$$\Pi_{IN}(\omega) = \frac{1}{2} \Re \int_S \hat{p}_{IN}(\mathbf{x}, 0^-, \omega) \overline{\hat{v}_{IN}(\mathbf{x}, 0^-, \omega)} \cdot \mathbf{e}_z dS,$$

and Π_T is the **transmitted power**:

$$\Pi_T(\omega) = \frac{1}{2} \Re \int_S \hat{p}_T(\mathbf{x}, 0^+, \omega) \overline{\hat{v}_T(\mathbf{x}, 0^+, \omega)} \cdot \mathbf{e}_z dS.$$

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- Identifying $\hat{p}_{\text{IN}}(\mathbf{x}, z, \omega) = P e^{i(\mathbf{k} \cdot \mathbf{x} + k_{1n} z)}$ and $\hat{p}_{\text{T}}(\mathbf{x}, z, \omega) = P T e^{i(\mathbf{k} \cdot \mathbf{x} + k_{2n} z)}$ one obtains with $-i\omega \varrho_j \hat{\mathbf{v}}_j \cdot \mathbf{e}_z = \partial_z \hat{p}_j$:

$$\Pi_{\text{IN}}(\omega) = \frac{k_{1n}(\mathbf{k}) P^2 S}{\varrho_1 \omega}, \quad \Pi_{\text{T}}(\omega) = \frac{\varrho_2 \omega^3 |W|^2 S}{k_{2n}(\mathbf{k})}.$$

- Consequently:

$$\begin{aligned} \tau(\omega) &= \frac{\varrho_1 \varrho_2 \omega^4}{k_{1n}(\mathbf{k}) k_{2n}(\mathbf{k})} \left| \frac{W}{P} \right|^2 \\ &= \frac{4 \varrho_1 \varrho_2 \omega^2}{k_{1n}(\mathbf{k}) k_{2n}(\mathbf{k})} \left| \mathcal{Z}_s(\omega) + \frac{i \varrho_1}{k_{1n}(\mathbf{k})} + \frac{i \varrho_2}{k_{2n}(\mathbf{k})} \right|^{-2} \end{aligned}$$

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- For $\rho_1 = \rho_2 = \rho_0$, $c_1 = c_2 = c_0$, and $\|\mathbf{k}\| = \frac{\omega}{c_0} \sin \theta$, and introducing the **coincidence frequency** $\omega_c = \frac{c_0^2}{\sin^2 \theta} \sqrt{\frac{\rho_s}{D}}$, one obtains:

$$\tau(\omega) = \frac{1}{1 + \left(\frac{\rho_s \omega \cos \theta}{2 \rho_0 c_0} \right)^2 \left(1 - \frac{\omega^2}{\omega_c^2} \right)^2}.$$

- The **transmission loss factor** (*indice d'affaiblissement*) TL is:

$$\begin{aligned} \text{TL}(\omega) &= 10 \log \left(\frac{1}{\tau(\omega)} \right) \\ &= 10 \log \left[1 + \left(\frac{\rho_s \omega \cos \theta}{2 \rho_0 c_0} \right)^2 \left(1 - \frac{\omega^2}{\omega_c^2} \right)^2 \right]. \end{aligned}$$

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- $TL = 0$ if $\Pi_T = \Pi_{IN}$, whereas $TL = \infty$ if $\Pi_T \ll \Pi_{IN}$.
- If $2\pi \ll \omega \ll \omega_c$, one obtains the **mass law**— TL depends on the mass of the plate solely:

$$TL \simeq 20 \log \left(\frac{\rho_s \cos \theta}{2\rho_0 c_0} \omega \right) .$$

- If $\omega = \omega_c$ then $TL = 0$ (but damping of the plate must be accounted for);
- If $\omega \gg \omega_c$, one obtains the **stiffness law**:

$$TL \simeq 60 \log(\alpha \omega) .$$

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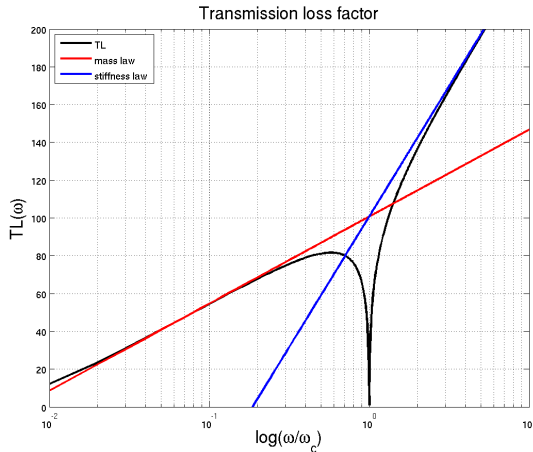
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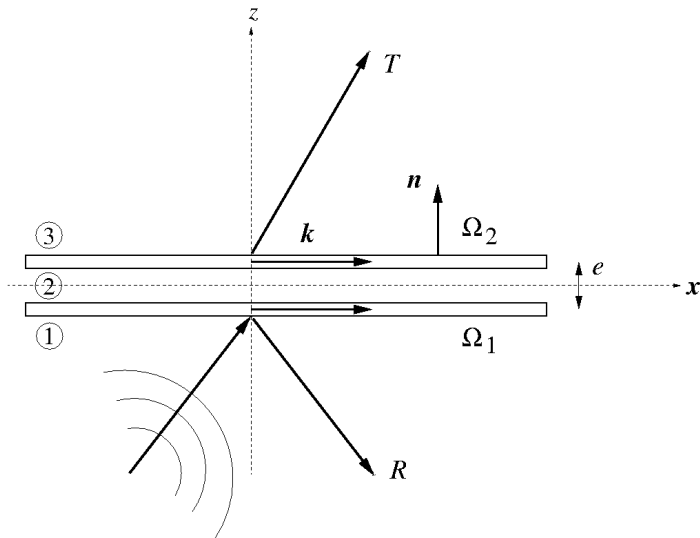
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- Fluid medium ①: incoming plane wave + reflected plane wave

$$\hat{p}_1(\mathbf{x}, z < 0, \omega) = P e^{i\mathbf{k} \cdot \mathbf{x}} \left(e^{ik_{1n}z} + R_1 e^{-ik_{1n}z} \right).$$

- Fluid medium ②: upward and downward plane waves

$$\hat{p}_2(\mathbf{x}, 0 < z < e, \omega) = P e^{i\mathbf{k} \cdot \mathbf{x}} \left(T_2 e^{ik_{2n}z} + R_2 e^{-ik_{2n}z} \right).$$

- Fluid medium ③: transmitted plane wave

$$\hat{p}_3(\mathbf{x}, z > e, \omega) = P T_3 e^{i(\mathbf{k} \cdot \mathbf{x} + k_{3n}z)}.$$

- R_1, R_2 and T_2, T_3 are the amplitude **reflection** and **transmission** coefficients, and $\mathbf{k}$ is the tangent wave vector.

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- The wave number in fluid medium ① is $k_1 = \frac{\omega}{c_1}$ and:

$$k_{1n}(\mathbf{k}) = k_1 \sqrt{1 - \left(\frac{\|\mathbf{k}\|}{k_1} \right)^2},$$

assuming that $\|\mathbf{k}\| < k_1$ for effective incident wave propagation.

- The wave number in fluid medium ②, $j = 2, 3$, is $k_j = \frac{\omega}{c_j}$ and:

$$k_{jn}(\mathbf{k}) = k_j \sqrt{1 - \left(\frac{\|\mathbf{k}\|}{k_j} \right)^2},$$

whereby transmission holds provided that $\|\mathbf{k}\| < k_j$
(total reflection otherwise).

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- Plates bending motions $\hat{w}_1(\mathbf{x}, \omega) = W_1 e^{i\mathbf{k} \cdot \mathbf{x}}$ and $\hat{w}_2(\mathbf{x}, \omega) = W_2 e^{i\mathbf{k} \cdot \mathbf{x}}$ such that:

$$\begin{aligned} -\omega^2 \mathcal{Z}_1(\omega) \hat{w}_1(\mathbf{x}, \omega) &= \hat{p}_1(\mathbf{x}, 0^-, \omega) - \hat{p}_2(\mathbf{x}, 0^+, \omega), \\ -\omega^2 \mathcal{Z}_2(\omega) \hat{w}_2(\mathbf{x}, \omega) &= \hat{p}_2(\mathbf{x}, e^-, \omega) - \hat{p}_3(\mathbf{x}, e^+, \omega), \end{aligned}$$

with boundary conditions (continuity of the normal speed $\varrho_j \omega^2 \hat{w}_1 = \partial_z \hat{p}_j|_{z=0^\pm}$ and $\varrho_j \omega^2 \hat{w}_2 = \partial_z \hat{p}_j|_{z=e^\pm}$)

$$\varrho_1 \omega^2 \hat{w}_1(\mathbf{x}, \omega) = i k_{1n}(\mathbf{k}) P (1 - R_1) e^{i\mathbf{k} \cdot \mathbf{x}},$$

$$\varrho_2 \omega^2 \hat{w}_1(\mathbf{x}, \omega) = i k_{2n}(\mathbf{k}) P (T_2 - R_2) e^{i\mathbf{k} \cdot \mathbf{x}},$$

$$\varrho_2 \omega^2 \hat{w}_2(\mathbf{x}, \omega) = i k_{2n}(\mathbf{k}) P (T_2 e^{ik_{2n}(\mathbf{k})e} - R_2 e^{-ik_{2n}(\mathbf{k})e}) e^{i\mathbf{k} \cdot \mathbf{x}},$$

$$\varrho_3 \omega^2 \hat{w}_2(\mathbf{x}, \omega) = i k_{3n}(\mathbf{k}) P T_3 e^{ik_{3n}(\mathbf{k})e} e^{i\mathbf{k} \cdot \mathbf{x}}.$$

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- One ends up with the system:

$$-\omega^2 \mathcal{Z}_1(\omega) W_1 = P(1 + R_1 - T_2 - R_2),$$

$$-\omega^2 \mathcal{Z}_2(\omega) W_2 = P(T_2 e^{ik_{2n}(\mathbf{k})e} + R_2 e^{-ik_{2n}(\mathbf{k})e} - T_3 e^{ik_{3n}(\mathbf{k})e}),$$

$$\varrho_1 \omega^2 W_1 = ik_{1n}(\mathbf{k}) P(1 - R_1),$$

$$\varrho_2 \omega^2 W_1 = ik_{2n}(\mathbf{k}) P(T_2 - R_2),$$

$$\varrho_2 \omega^2 W_2 = ik_{2n}(\mathbf{k}) P(T_2 e^{ik_{2n}(\mathbf{k})e} - R_2 e^{-ik_{2n}(\mathbf{k})e}),$$

$$\varrho_3 \omega^2 W_2 = ik_{3n}(\mathbf{k}) P T_3 e^{ik_{3n}(\mathbf{k})e}.$$

- From the previous computation of the incident and transmitted powers:

$$\text{TL}(\omega) = -10 \log \left(\frac{\varrho_1 \varrho_3 \omega^4}{k_{1n}(\mathbf{k}) k_{3n}(\mathbf{k})} \left| \frac{W_2}{P} \right|^2 \right) = 20 \log \left(\frac{\cos \theta}{\varrho_0 c_0 \omega} \left| \frac{P}{W_2} \right| \right)$$

whenever $c_1 = c_2 = c_3 = c_0$ and $\varrho_1 = \varrho_2 = \varrho_3 = \varrho_0$
(speed of sound and density of air typically).

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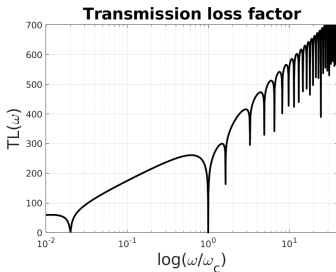
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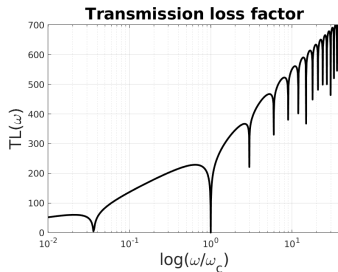
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(i) $\theta = \frac{\pi}{8}$



(ii) $\theta = \frac{\pi}{6}$

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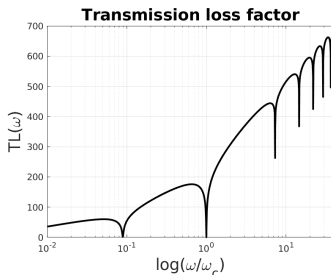
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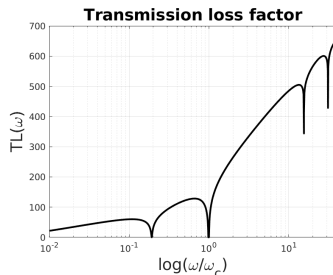
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(iii) $\theta = \frac{\pi}{4}$



(iv) $\theta = \frac{\pi}{3}$

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